

Question Number	Answer	Additional Guidance	Mark
2(a)(i)	A description that makes reference to the following points: - glucose and fructose (1) - joined by glycosidic bond (1)		(2)

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2(a)(ii)	pH or Ph or PH	ignore concentration, level or any other extra to pH. ACCEPT acidity, alkalinity	(1)

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2(a)(iii)	An answer that makes reference to the following points: - temperature affects {action / reaction rate / number of (S and E) collisions} of enzyme (1) (25°C) {may be / is} optimum (for invertase) / gives rate which is not too fast or slow / (1)	ACCEPT so that only pH affects rate ACCEPT above 25°C denatures enzyme	(2)

Question Number	Answer	Additional Guidance	Mark
2(a)(iv)	invertase / sucrose concentration (1)	ACCEPT substrate / enzyme	(1)

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2(a)(v)	A description that makes reference to the following points: - dissolve a stated mass of {invertase / sucrose} in a stated volume of water / use a stock solution of {sucrose / invertase} (1) - use the same volume of this solution for each pH (1)	ecf from aiv	(2)

Question Number	Answer	Additional Guidance	Mark
2(a)(vi)	An explanation that makes reference to the following points: - the rate slows down as {sucrose is used up / sucrose becomes limiting} (1) - so (rates / results) cannot be compared (1)	ACCEPT converse argument ACCEPT substrate ACCEPT converse argument	(2)

Question Number	Answer	Additional Guidance	Mark
2(a)(vii)	An answer that makes reference to the following points: • measure the mass of {sucrose broken down / products formed} at intervals over time (1) • plot a graph of mass (y) against time (x) (1) • find the gradient of the graph at its start (1)	ACCEPT concentration / quantity / amount	(3)

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2(b)(i)	A table showing the following features: • suitable table drawn (1) • headings of pH and Initial rate of reaction as a percentage of the maximum initial rate (1) • all data correctly entered (1)	example table <table><tr><th>pH</th><th>Initial rate of reaction as a percentage of the maximum initial rate</th></tr><tr><td>3.5</td><td>45</td></tr><tr><td>4.5</td><td>84</td></tr><tr><td>5.5</td><td>100</td></tr><tr><td>6.5</td><td>90</td></tr><tr><td>7.5</td><td>74</td></tr><tr><td>8.5</td><td>33</td></tr></table>	pH	Initial rate of reaction as a percentage of the maximum initial rate	3.5	45	4.5	84	5.5	100	6.5	90	7.5	74	8.5	33	(3)
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2(b)(ii)	An answer which includes the following points: • (from these results, the conclusion is valid as) the highest initial rate measured is at pH 5.5 (1) • measurements at (smaller) intervals {were not made / should have been made} (1) • optimum might be anywhere (else) between 4.5 and 6.5 (1)	ACCEPT converse argument	(3)

Question Number	Answer	Additional Guidance	Mark
2(c)	An answer which includes at least one similarity and one difference: Similarities: • both show an optimum (1) • both have same / similar activity at 70 C (1) Differences • optimum is at a lower temperature for immobilised than non-immobilised (1) • temperature affects immobilised enzymes more (1)	ACCEPT both rise to a point and then fall ACCEPT converse, correct figures quoted ACCEPT converse	(3)